

Bone remodeling is the continuous, microscopic process by which bone tissue throughout the body is resorbed and redeposited. Proper regulation of bone remodeling is necessary to maintain bone strength and prevent the onset of fragile bone conditions such as osteoporosis. Patients with such conditions have increased fracture risk and generally exhibit decreased osteoblast (bone-depositing cell) activity, leading to reduced bone density and weakening of the bone matrix. However, some cases can occur due to overactive osteoclasts (multinucleated cells that mediate bone resorption). Osteoclasts resorb bone by secreting proteolytic enzymes and acid into the adjacent extracellular space. These enzymes degrade the organic matrix of bone, and the acid causes the dissolution and release of the mineral components of bone such as calcium, potassium, and magnesium.

Osteoclast differentiation and activation is thought to be related to the RANKL/RANK signaling pathway. RANKL, expressed as a membrane-bound or secreted protein, serves as the ligand for RANK, a transmembrane receptor. In bone tissue, RANK is expressed by precursors of mature osteoclasts, also known as osteoclast progenitor cells (OPCs). RANKL can be expressed by osteoblasts adjacent to OPCs, and the binding of RANKL to RANK in these neighboring cells causes downstream effects through recruitment of TRAF6, an intracellular protein. RANK is involved in multiple downstream signaling cascades, and four of these directly mediate the development of mature osteoclasts from OPCs. In one of the four pathways, the transcription factor NF- κ B is activated, which leads to increased expression of the proto-oncogene c-Fos. Subsequently, c-Fos interacts with another transcription factor known as NFATc1 to promote the expression of genes related to osteoclast-specific development and function in OPCs (Figure 1).

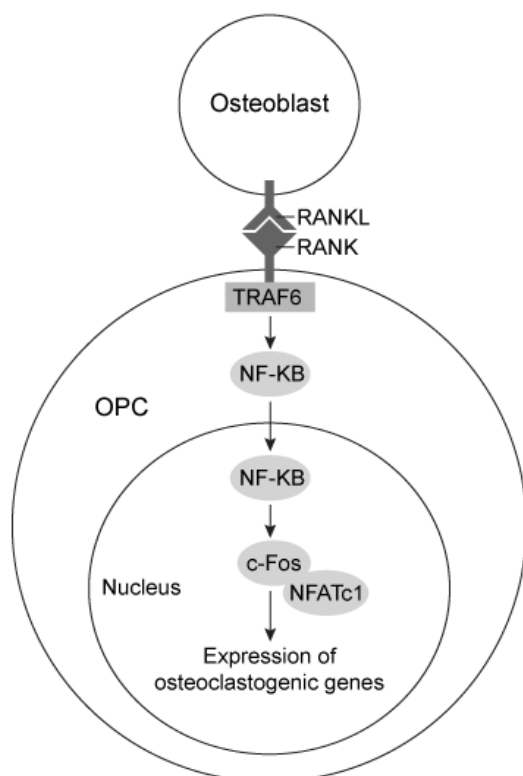


Figure 1 RANKL/RANK signaling induces differentiation of OPCs into osteoclasts

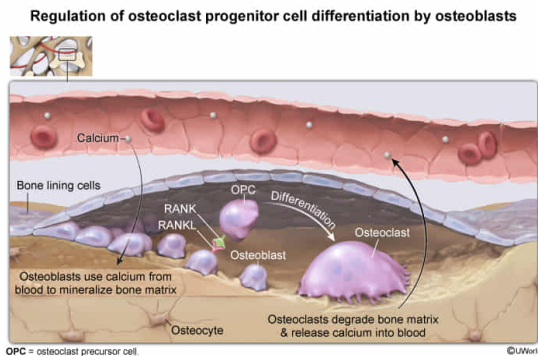
Based on the passage, the cells that directly stimulate OPCs to differentiate are cells that promote:

- ☐ A. calcium transfer from bone tissue into the bloodstream. [36%]
- ✓ ☒ B. calcium transfer from the bloodstream into bone tissue. [42%]
- ☐ C. acid production in bone tissue. [2%]
- ☐ D. proteolytic enzyme production in bone tissue. [18%]

Correct

42% answered correctly

Explanation:



Bone is a type of **connective tissue** that forms the vertebrate **endoskeleton**. The bone matrix consists of nonliving organic materials (eg, collagen) and inorganic materials (eg, calcium phosphate). In addition, several cell types in bone (osteocytes, osteoclasts, osteoblasts) function to maintain the strength and integrity of the skeleton through **bone remodeling**. During bone remodeling, osteoclasts break down (resorb) old bone and **osteoblasts deposit new bone**. Specifically, osteoblasts *promote* the precipitation of **calcium and phosphate** from the bloodstream and their transfer and incorporation into the bone matrix.

The passage states that **RANKL** is a ligand **expressed by osteoblasts** that binds RANK, a transmembrane receptor expressed by osteoclast progenitor cells (OPCs). The binding of RANKL on osteoblasts to RANK on OPCs induces downstream signaling pathways that lead to the differentiation of OPCs into mature osteoclasts. Accordingly, **osteoblasts**, which **promote the absorption of calcium from blood to bone tissue**, directly stimulate OPCs to differentiate.

(Choices A, C, and D) According to the passage, *osteoclasts* secrete proteolytic enzymes and acid that break down the organic and mineral components of bone. As the bone matrix is broken down, mineral components (eg, calcium) are reabsorbed from the bone into the bloodstream. However, RANKL, the ligand that stimulates OPC differentiation, is expressed by *bone-depositing* osteoblasts, *not* osteoclasts that promote the breakdown of bone.

Educational objective:

Bone remodeling is a continuous process in which osteoclasts (bone-resorbing cells) break down old bone and osteoblasts (bone-depositing cells) secrete new bone matrix. This process functions to maintain the strength and integrity of bone over time. Osteoblasts promote the precipitation of calcium and phosphate from the bloodstream and their transfer and incorporation into the bone matrix. In contrast, osteoclasts secrete acid proteolytic enzymes and acid that break down the mineral components of bone, releasing calcium and phosphate into the bloodstream.

Time spent:0 sec

Subject:
Biology

QID:401899

System:
Musculoskeletal System

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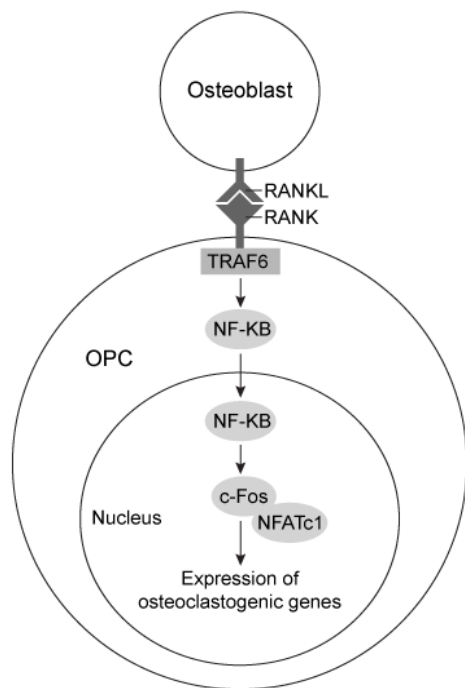


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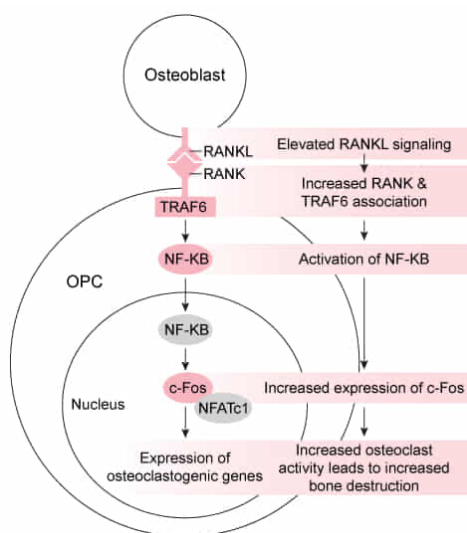
A group of scientists hypothesized that elevated RANKL signaling contributes to the decrease in bone density associated with osteoporosis. To test their hypothesis, the scientists collected bone tissue samples from patients with the disease and compared them to healthy controls. Based on this information, all of the following experimental observations would support their hypothesis EXCEPT:

- ☐ A. higher nuclear c-Fos expression in patient samples than in control samples. [7%]
- ☐ B. more RANK/TRAF6 association in patient samples than in control samples. [18%]
- ☐ C. increased RANKL/RANK binding in patient samples than in control samples. [16%]
- ☒ D. higher levels of cytoplasmic NFATc1 in patient samples than in control samples. [57%]

Correct

57% answered correctly

Explanation:



In healthy individuals, the process of bone remodeling is tightly regulated to ensure that bone mass and density remain constant (ie, the rate of bone resorption by osteoclasts equals the rate of bone deposition by osteoblasts). According to the passage, conditions that predispose patients to fragile bones generally *decrease* osteoblast activity or *increase* osteoclast activity and thereby decrease bone density and weaken the bone matrix.

In this experiment, scientists compared bone tissue samples from patients with osteoporosis to samples from healthy controls. They hypothesized that elevated RANKL signaling contributes to declining bone density in these patients. The passage states that osteoclasts resorb bone and that osteoclast differentiation and activity are regulated by the RANKL/RANK signaling pathway. Accordingly, if the scientists' hypothesis is true, increased RANKL signaling in osteoporosis patients compared to healthy controls would likely induce the following:

- *Higher nuclear c-Fos expression.* The passage states that c-Fos expression is increased via the actions of NF-κB, a transcription factor *activated* by RANKL/RANK signaling. This c-Fos then interacts with NFATc1 (another transcription factor) in the *nucleus* to increase expression of genes important for osteoclast function. Therefore, increased nuclear c-Fos expression would indicate increased RANKL activity in these patients (**Choice A**).
- *More association of RANK and TRAF6.* According to the passage, RANK activation by RANKL recruits some of the cytosolic TRAF6 molecules to the RANKL/RANK signaling pathway. Therefore, increased RANKL signaling would most likely *increase* the association of TRAF6 and RANK (**Choice B**).
- *Increased binding of RANKL to RANK.* Based on the passage, RANKL binds RANK to induce downstream signals that ultimately lead to osteoclast differentiation and activation (**Choice C**).

Based on the passage, activation of the RANKL/RANK cascade in OPCs leads to increased expression of c-Fos, which then interacts with NFATc1 in the nucleus to induce expression of genes necessary for osteoclast activation. Because NFATc1 functions in the *nucleus* (*not* the cytoplasm) during RANKL/RANK signaling, higher levels of *cytoplasmic* NFATc1 in patient samples would *not* necessarily support the scientists' hypothesis.

Educational objective:

Decreased activity of osteoblasts (bone-depositing cells) or increased activity of osteoclasts (bone-resorbing cells) can lead to reduced bone mass and density.

Subject:
Biology

System:
Musculoskeletal System

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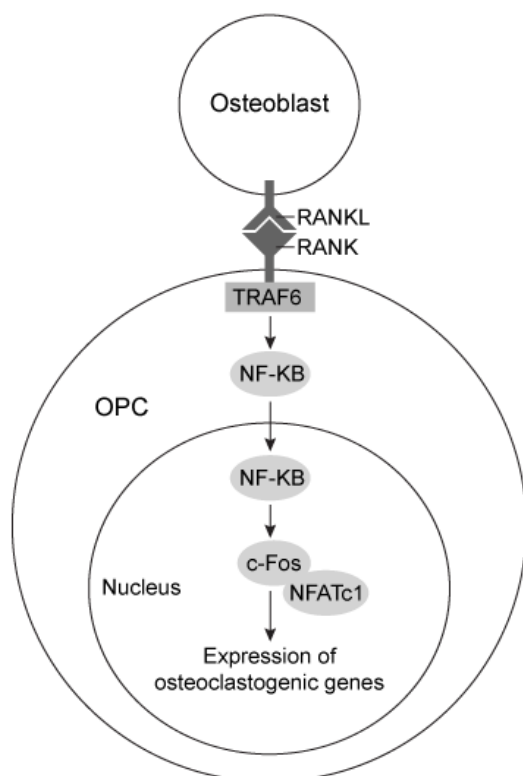
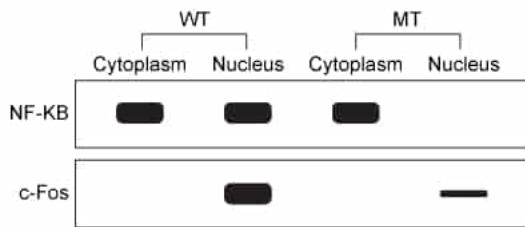


Figure 1 RANKL/RANK signaling induces differentiation of OPCs into osteoclasts

A student isolates the cellular components of OPCs in wild-type (WT) mice and in mutant (MT) mice exhibiting a single base substitution that causes reduced osteoclast activity. Western blot

analysis is then performed to test for NF- κ B and c-Fos expression in the cytoplasm and nucleus of OPCs isolated from these mice. The results of this analysis are shown below.



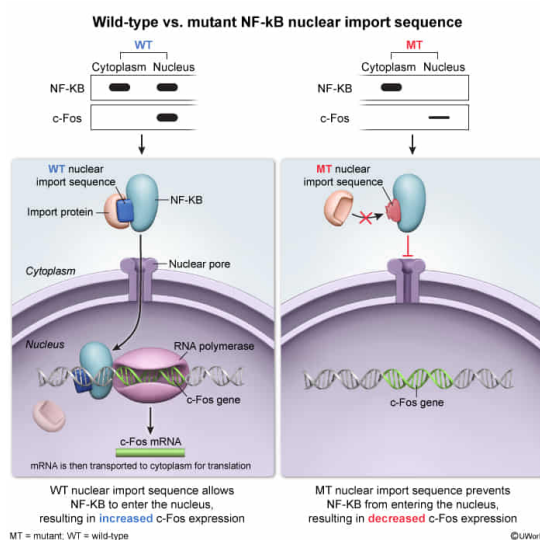
Based on the passage, which statement best explains these results?

- ☒ A. Mutant mice have a mutation that disables the ability of NF- κ B to localize to the nucleus. [74%]
- ☐ B. Mutant mice have a mutation that prevents c-Fos from interacting with NF- κ B. [8%]
- ☐ C. Mutant mice have a mutation in the DNA binding region of NF- κ B. [10%]
- ☐ D. Mutant mice have a mutation that prevents the transcription of c-Fos in osteoclast progenitor cells. [7%]

Correct

73% answered correctly

Explanation:



Transcription factors are proteins that can alter gene expression by binding DNA near gene promoter regions to either increase or decrease the transcription of a target gene. Like all proteins, transcription factors are translated in the cytoplasm and must enter the nucleus (where DNA is stored) to influence gene expression. Transcription factors rely on their **nuclear localization sequence** to gain entrance to the nucleus. The nuclear localization sequence is composed of amino acids and functions as a nuclear import tag that is recognized and bound by nuclear import proteins, which facilitate the transport of transcription factors into the nucleus.

According to the passage, RANK signaling activates the transcription factor NF- κ B. Upon activation, NF- κ B enters the nucleus and acts to increase c-Fos gene expression. In this scenario, a student analyzes the differences in OPCs from wild-type (WT) mice and mutant (MT) mice with reduced osteoclast activity. Following isolation of cellular components, **western**

blotting is used to test for NF- κ B and c-Fos expression in the nucleus and cytoplasm of WT and MT OPCs. Western blot results show that, compared to WT mice, MT mice *lack* nuclear expression of NF- κ B and exhibit *reduced* c-Fos expression. Because NF- κ B requires an intact *nuclear localization sequence* to enter the nucleus and increase c-Fos expression, a mutation in the NF- κ B nuclear localization sequence of MT mice best explains these results.

(Choice B) According to the passage, NF- κ B increases the expression of c-Fos but does *not* interact directly with the c-Fos protein. However, c-Fos does interact with NFATc1 to induce the expression of osteoclastogenic genes.

(Choice C) Transcription factors contain DNA binding regions that allow them to interact with DNA in the nucleus. A mutation in the DNA binding region of NF- κ B would prevent it from interacting with DNA and increasing the expression of c-Fos. However, a mutation in this region would *not* prevent NF- κ B from entering the nucleus.

(Choice D) Based on the western blot, researchers still detected c-Fos in the nucleus of MT mice, indicating that c-Fos was still *transcribed* (ie, expressed) in these mice but at *reduced* levels.

Educational objective:

Transcription factors have nuclear localization sequences that facilitate their entry into the cell nucleus, where they regulate the expression of target genes.

Time spent:0 sec

Subject:
Biology

QID:401901

System:
Musculoskeletal System

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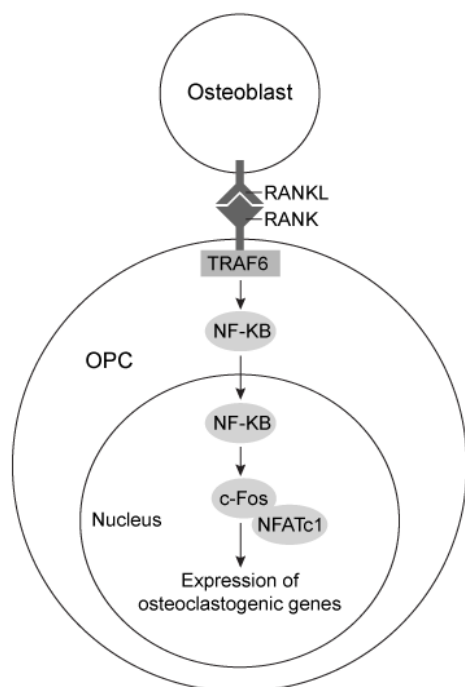


Figure 1 RANKL/RANK signaling induces differentiation of OPCs into osteoclasts

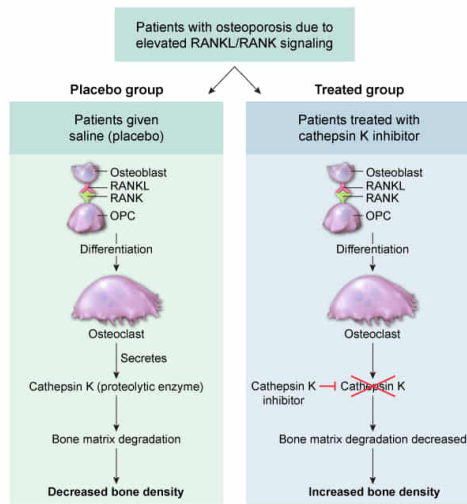
Cathepsin K, a proteolytic enzyme secreted by osteoclasts, has a key role in bone remodeling. In a clinical trial, patients with osteoporosis due to increased RANKL/RANK signaling were assigned to either an untreated group (saline placebo administered) or a treated group (cathepsin K inhibitor administered). Given the information in the passage, bone density after the trial was most likely:

- ☐ A. increased in the untreated group compared to the treated group. [15%]
- ✓ ☒ B. decreased in the untreated group compared to the treated group. [67%]
- ☐ C. nearly the same in both groups. [6%]
- ☐ D. nearly the same in both groups initially but then increased in the untreated group compared to the treated group. [10%]

Correct

68% answered correctly

Explanation:



According to the passage, some cases of osteoporosis (a fragile bone condition) arise due to overactive osteoclast activity. The passage states that osteoclasts are cells that break down (resorb) bone and that binding of RANKL to RANK induces osteoclast progenitor cells (OPCs) to differentiate into mature osteoclasts. These mature osteoclasts secrete acid that demineralizes bone; they also express osteoclast-specific genes such as those encoding the **proteolytic enzymes** (eg, cathepsin K) responsible for degrading the organic bone matrix.

In this scenario, patients with *osteoporosis* due to *increased RANKL/RANK signaling* were assigned to the following two groups during a clinical trial:

- An untreated group, which received a saline placebo
- A treated group, which received a drug that inhibits cathepsin K, a proteolytic enzyme secreted by osteoclasts that *breaks down* the bone matrix

When untreated patients were given the saline (placebo), the effects of their heightened RANKL/RANK signaling most likely continued unchanged. As a result, RANKL/RANK-induced OPC differentiation into osteoclasts likely continued in untreated patients. This likely allowed osteoclasts to increase in number and secrete cathepsin K to break down bone, further *decreasing* bone density in the untreated patients.

In contrast, when patients in the treated group were given the cathepsin K inhibitor, this inhibitor likely *prevented* cathepsin K from breaking down bone, *increasing* the patients' bone density over time (ie, via slowing bone density loss). Accordingly, cathepsin K activity was likely decreased in patients treated with the inhibitor but *remained* overactive in untreated patients due to the underlying increased RANKL/RANK signaling. As a result, *after the trial*, **bone density** was most likely **decreased in the untreated group** compared to the treated group (**Choices A, C, and D**).

Educational objective:

During bone resorption, osteoclasts release proteolytic enzymes that break down the organic bone matrix. In patients with osteoporosis, overactivity of osteoclasts and increased release of these enzymes can lead to decreased bone density over time.

Time spent: 0 sec
Subject:
Biology

QID: 401902
System:
Musculoskeletal System